Project Report On

Personalized Anomaly Detection in Financial Transactions

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

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CERTIFICATE

This is to certify that the project report entitled ***"Personalized Anomaly Detection in Financial Transactions"*** *is a bonafide record of the work done by* ***Aaron James (U2203003), Abhishek S (U2203011), Augustin Vincent (U2203061) and Bosco Reji Jose (U2203069)****, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

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Abstract

Financial fraud detection has become a major challenge in modern digital economies due to the increasing scale and complexity of online transactions. Traditional systems rely on global behavioral patterns, often failing to capture individual user differences, which leads to false positives and missed fraud cases.

This project proposes a personalized anomaly detection framework , Account GANomaly, which uses a user-conditioned Generative Adversarial Network to model individual transaction behavior. It incorporates a Hybrid User Encoder that combines short-term transaction patterns using Long Short-Term Memory networks and long-term behavioral statistics using a Multilayer Perceptron to generate robust user embeddings. These embeddings are integrated into the GAN architecture to enable user-specific transaction reconstruction.

Anomaly scores are calculated based on reconstruction error and latent space differences, and further refined using an XGBoost classifier. The model is evaluated on the IEEE-CIS fraud detection dataset and synthetic data, showing improved detection accuracy and reduced false positives, making it an effective and scalable solution for real-time fraud detection.

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List of Abbreviations

AI: Artificial Intelligence

ALGAN: Adjusted-LSTM GAN

AUPRC: Area Under the Precision-Recall Curve

GAN: Generative Adversarial Network

GANomaly: A specific anomaly detection architecture using an encoder-decoder-encoder pipeline

LSTM: Long Short-Term Memory

MLP: Multilayer Perceptron

VAE: Variational Autoencoder

XGBoost: Extreme Gradient Boosting

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Chapter 1

Introduction

This project introduces Account GANomaly, a novel, user-conditioned Generative Adversarial Network (GANomaly) designed for personalized anomaly scoring in financial transactions. The system is engineered to move beyond global fraud detection rules by learning the unique behavioral profile of each user, thereby providing a more accurate and robust measure of transactional irregularity. The core objective is to significantly enhance the efficiency and accuracy of fraud detection by delivering a personalized anomaly score for every transaction.

1.1 Background

Financial fraud and anomaly detection within financial transactions are considered critical tasks. The landscape of finance has become increasingly complex due to the growth of outsourcing, leading to intricate transaction networks where fraudulent activities can be harder to detect. Traditional reliance on manual checks and auditing is often inadequate, as human auditors cannot efficiently process the millions of transactions generated, resulting in delayed detection and increased risk of financial losses. Advancements in AI and modern machine learning algorithms, such as Isolation Forests and Autoencoders, have been proposed to detect suspicious patterns more quickly and accurately than manual methods. The central issue that current methods struggle with is the context of user-specific behavior. A transaction considered "normal" for one user, such as a high-volume spender, can be considered highly anomalous for another. This necessitates a novel approach that can learn the personalized baseline of normality for each individual user.

1.2 Problem Definition

The core problem is that large volumes of complex financial transactions overwhelm traditional auditing systems. Subtle anomalies are often missed by simple rule-based approaches, leading to delayed fraud detection and increased risk of financial misstatements. A key challenge is that "normal" behavior varies across users and changes over time, making it difficult to distinguish legitimate high-value transactions from fraudulent ones, especially with sparse data. To address this, the problem is framed as training an unsupervised model that reconstructs transactions accurately only when they match a user's profile (u), using reconstruction error as a measure of personalized anomalies. This highlights the need for a unified, intelligent framework that combines deep personalization with automated efficiency.

1.3 Scope and Motivation

Scope: The current scope of the Account GANomaly project is focused on the successful design and validation of its core personalized detection mechanism: detecting suspicious transactions in historical data using a Hybrid User Encoder to generate a robust User Embedding. This embedding will be used to condition the GANomaly architecture to achieve user-specific reconstruction and output a personalized anomaly score. While the project is intended to lay the foundation for future real-time monitoring and integration with fraud prevention tools, the current scope does not include full, production-ready real-time API deployment.

Motivation: The motivation for the Account GANomaly project stems from the inadequacy of traditional fraud detection methods, which fail to capture user-specific behavior in complex, high-volume financial transactions. With increasing financial fraud and outsourcing leading to complex transaction networks, transactions deemed "normal" for one user (a high-volume spender) can be highly anomalous for another, proving that the definition of "normal" is both non-stationary and user-dependent. The project is motivated by the critical need for a model that learns a personalized baseline of normality for each user, which will significantly reduce financial losses and improve the efficiency and accuracy of anomaly detection compared to traditional auditing processes.

1.4 Objectives

The primary goal of this project is to create an effective and personalized solution for anomaly detection in financial transactions. The following specific, measurable, achievable, relevant, and time-bound objectives guide the development and validation of the Account GANomaly system:

- To develop a user-conditioned Generative Adversarial Network (GAN) coupled with an XGBoost classifier to establish a personalized baseline for every account.
- To integrate short-term transaction sequences and long-term aggregated statistics into a robust user embedding to identify profile-unique deviations
- To train the conditioned Generative Adversarial Network to minimize the Reconstruction Error for only "normal" transactions unique to the user profile, making the final error score a direct measure of personalized anomaly.
- To maximize detection accuracy and reduce false positives by ensuring the model only reconstructs features aligning with the learned behavior of a particular user.

1.5 Challenges

Developing a personalized, AI-based anomaly detection system like Account GANomaly presents several significant technical and operational hurdles. We have identified several key risks and technical hurdles that will require careful consideration:

- **Extreme Data Sparsity:** Many users have very few transactions in the dataset . Because of this, models like LSTM and MLP struggle to learn a reliable behavioral pattern for each user.Without enough data, the system cannot build a strong user embedding, which reduces anomaly detection accuracy.
- **Adversarial Training Instability:** The system uses Generative Adversarial Networks (GANs) with a Generator and Discriminator.Training these two networks simultaneously is difficult because the generator may dominate the discriminator,mode collapse may occur (generator produces limited outputs), gradient disappearance may happen during training.This makes the training process unstable and requires careful tuning.

- **Cold-Start Problem:** New users or accounts do not have historical transaction data.The system cannot immediately learn their normal behavior.This may lead to high false positives in the beginning until enough transactions are collected.
- **Concept Drift:** User financial behavior changes over time due to , lifestyle changes , seasonal spending and new purchasing habits.If the model is not updated frequently, it may treat normal behavior changes as fraud, increasing false alarms.

1.6 Assumptions

The successful execution of this project is based on the following assumptions:

- It is assumed that even though financial behavior changes over time, users still follow certain underlying patterns.
- The model assumes that the transaction data is accurate and reliable.
- The system assumes that fraudulent transactions will significantly deviate from a user's normal behavior.
- The model assumes that changes in user behavior happen gradually, not suddenly.

1.7 Societal / Industrial Relevance

The AI-Based Anomaly Detection project holds significant relevance for both the financial industry and the wider society, primarily by enhancing fraud prevention and increasing trust in financial systems.

Industrial Relevance: The project addresses critical pain points faced by Financial Institutions, Accounting firms, Auditors, and even Small Businesses. The use of GANs (Generative Adversarial Networks) and Autoencoders enables the processing of large volumes of complex financial data efficiently, directly overcoming the limitations of manual auditing and traditional rule-based systems. The novelty of the Account GANomaly system—integrating a behavioral fingerprint and personalized flagging—allows for the identification of both overt and subtle anomalies and hidden irregularities that occur within a user-specific context, which are frequently missed by non-personalized methods. By achieving early anomaly detection, the system helps organizations reduce financial losses,

prevent the escalation of fraudulent activities, and build trust in the accuracy and integrity of financial reporting. The system architecture lays the groundwork for future real-time monitoring and integration with existing fraud prevention tools, enabling proactive risk management.

Societal Relevance: For the broader society, the successful deployment of this technology translates directly to a more secure financial environment. More accurate and timely fraud detection protects consumers, businesses, and the financial system from the destabilizing effects of financial crime and identity theft. Improved detection accuracy and consistency enhance transparency and reinforce public confidence in the financial sector's ability to safeguard assets and data.

1.8 Organization of the Report

This report is organized into four chapters, detailing the foundation, design, and implementation plan for the Account GANomaly system, which focuses on the detection of anomalies in financial transactions.

Chapter 1 Introduction: This chapter defines the problem of user-dependent "normal" transactions. Introduces the novelty of user-conditioned GANomaly and states the core objective of learning a personalized baseline of normality.

Chapter 2 Literature Survey: This chapter reviews existing methods (GANs, VAEs, LSTMs) and identifies the critical gap: the lack of robust personalized user behavior modeling. Discusses the foundation of the original GANomaly architecture.

Chapter 3 System Design: This chapter specifies requirements like Real-Time Scoring (under 50ms) and Scalability. Presents the core architecture, detailing the inputs (x_t, seq_recent, $profile_stats$) and the design of the Hybrid User Encoder (LSTM/GRU + MLP fusion) for generating the User Embedding (u).

Chapter 4 Implementation: This chapter describes the practical implementation of the proposed Account-GANomaly framework for personalized anomaly detection in

financial transactions. The goal of the implementation is to develop a hybrid system capable of learning individual user behavior patterns and identifying fraudulent transactions by detecting deviations from these personalized baselines.

Chapter 5 Results and Discussions: This section presents the experimental results of the proposed Account-GANomaly frame- work and analyzes its performance in detecting fraudulent financial transactions. The model was evaluated using two datasets: the IEEE-CIS Fraud Detection dataset and a synthetic transaction dataset designed to capture behavioral transaction patterns.

Chapter 6 Conclusions Future Scope: This chapter summarizes the achievement of developing the user-conditioned GANomaly framework. Reiterates the success metric (improved AUPRC) and discusses future enhancements.

1.9 Summary of Chapter

This Chapter introduces Account GANomaly, a novel, user-conditioned Generative Adversarial Network (GANomaly) designed for personalized anomaly scoring in financial transactions. It establishes the problem of traditional fraud detection methods failing due to the high volume and user-dependent, non-stationary nature of "normal" transaction behavior. The core objective is to move beyond global rules by learning a personalized baseline of normality for each user via a Hybrid User Encoder and conditioning the GANomaly architecture, making the reconstruction error a direct measure of personalized anomaly. The chapter outlines the specific goals, including designing the encoder and achieving superior performance, while also identifying key challenges like GAN training instability and concept drift. The project is motivated by the critical need to significantly enhance the accuracy and efficiency of fraud detection, which holds high relevance for the financial industry and society by reducing losses and increasing public trust.

Chapter 2

Literature Survey

2.1 Introduction

The contemporary financial landscape is characterized by high transaction volumes, evolving fraud schemes, and a complex interplay of user-specific behavioral patterns, making anomaly detection a persistent challenge. Traditional methods, often reliant on static rules or simple statistical models, frequently fail to capture the subtle, non-stationary nature of "normal" behavior, especially when individual user contexts are involved. The advent of advanced deep learning architectures, particularly those leveraging Generative Adversarial Networks (GANs) and recurrent neural networks, presents a transformative pathway to address these complexities. Architectures focused on feature learning, sequence modeling, and distributional analysis are increasingly applied to detect outliers and structural abnormalities in financial time series and transaction networks. This chapter surveys seminal and contemporary research, analyzing key methodologies in anomaly detection to establish the foundational principles and highlight the critical research gap addressed by the proposed Account GANomaly system.

2.2 Survey Papers

2.2.1 Utilizing GANs for Fraud Detection: Model Training with Synthetic Transaction Data [1]

- **Methodology:** Uses Generative Adversarial Networks (GANs) to detect fraudulent transactions.GAN consists of:Generator (G) → generates synthetic transaction data.Discriminator (D) → distinguishes real vs fake transactions.The system trains on normal transaction data distribution.Fraud detection is performed by comparing reconstructed normal data with real transactions.Synthetic data is used to improve

model training and robustness.

- **Advantages:** It can handle imbalanced datasets where fraud samples are rare . Generates synthetic data to improve training quality.Detects complex fraud patterns using deep learning.Improves transaction security systems.

- **Relevance to Project:** Highly relevant for projects involving: Fraud detection , Financial anomaly detection , Cybersecurity.Helps detect bot-generated or suspicious transactions.Improves system security and reliability.

2.2.2 GANomaly: Semi-Supervised Anomaly Detection via Adversarial Training [2]

- **Methodology:** It uses a Generative Adversarial Network (GAN) architecture for anomaly detection.The generator consists of an Encoder → Decoder → Encoder pipeline.The first encoder converts the input image into a latent vector representation.The decoder reconstructs the image from the latent vector.The second encoder maps the reconstructed image back to latent space.During training, the model learns only normal samples and tries to reconstruct them accurately.During testing, anomaly score is calculated using reconstruction error and latent space difference.If the difference between original and reconstructed representation is high → anomaly detected.

- **Advantages:** It works with semi-supervised learning (needs only normal data). Captures complex data distributions using adversarial training.More computationally efficient compared to earlier GAN-based anomaly methods.Learns features in both image space and latent space.Detects unknown or unseen anomalies effectively.

- **Relevance to Project:** Suitable for AI-based anomaly detection systems.Helps identify abnormal patterns in datasets where anomalous samples are rare.Can be used in:Fraud detection , Security systems , IoT monitoring , Driver behaviour detection.Provides a deep learning-based approach for detecting abnormal behaviour.

2.2.3 XGBoost: A Scalable Tree Boosting System [3]

- **Methodology:** It uses gradient boosted decision trees.Builds models iteratively where each new tree corrects errors of the previous one.Uses:Regularized objective function , Parallel computing and Sparse data handling.

- **Advantages:** Very high prediction accuracy.Fast and scalable for large datasets. Handles missing data and sparse datasets well.Widely used in real-world machine learning competitions.

- **Relevance to Project:** Can be used for:Classification of normal vs anomaly data , Fraud detection , Risk prediction , Works well with structured datasets like transaction records.

2.2.4 ALGAN:Time Series Anomaly Detection with Adjusted- LSTM GAN [4]

- **Methodology:** Introduces a GAN model combined with Adjusted-LSTM (ALstm) for anomaly detection in time-series data.The Architecture includes:Generator , Discriminator , Adjusted-LSTM layer with attention mechanism.The ALstm adjusts LSTM outputs using attention over hidden states and inputs.The model learns temporal patterns in time-series data.An anomaly score is computed from the difference between original and reconstructed data.

- **Advantages:** Captures long-term temporal dependencies better than traditional LSTM.Works for both univariate and multivariate time-series datasets.Reduces information loss in long sequences.Performs better than traditional methods like: PCA , Isolation Forest and Autoencoders.

- **Relevance to Project:** Useful if the project deals with time-series data such as: Sensor data , Network traffic , Transaction logs.Can detect suspicious or abnormal activity over time. Helps improve real-time monitoring systems.

2.2.5 FraudNE: Joint Embedding Approach for Fraud Detection [5]

- **Methodology:** Uses a Generative Adversarial Network (GAN) framework for anomaly detection.The generator contains an Encoder → Decoder → Encoder archi-

tecture.The first encoder converts the input image into a latent vector representation.The decoder reconstructs the image from the latent vector.The second encoder maps the reconstructed image back into latent space.During training, the model learns normal data distribution.During testing, an anomaly score is computed using reconstruction error and latent space difference.If the difference is high → the sample is classified as an anomaly.

- **Advantages:** Requires only normal data for training (semi-supervised).Detects unknown or unseen anomalies.Learns both image features and latent representations.Better accuracy and efficiency compared with earlier GAN models.

- **Relevance to Project:**Useful for detecting abnormal patterns in datasets.Can be applied to:Fraud detection , Network intrusion detection and System anomaly monitoring.Helps build AI-based anomaly detection systems.

2.3 Comparison and Gaps Identified

Table 2.1: Comparative Analysis of Surveyed Literature

Paper Title	**Methodology**	**Advantages**	**Disadvantages**
Utilizing GANs for Fraud Detection: Model Training with Synthetic Transaction Data	Uses Generative Adversarial Networks (GANs) to generate synthetic transaction data and improve fraud detection models.	Unsupervised learning capability	Requires large training data
GANomaly: Semi-Supervised Anomaly Detection via Adversarial Training	Uses Conditional GAN (cGAN) architecture to learn normal data patterns and detect anomalies through reconstruction errors.	Works well with imbalanced datasets	Computationally expensive
XGBoost: A Scalable Tree Boosting System & Distributional Anomalies	Uses gradient boosting decision trees with regularized objective functions for classification and prediction.	Efficient handling of sparse data	Limited improvements when over-ensemble is used
ALGAN: Time Series Anomaly Detection with Adjusted LSTM GAN	Combines LSTM with GAN to detect anomalies in time-series data by learning sequential patterns.	Strong performance in time-series datasets	Slower training due to sequential modeling
FraudNE: Joint Embedding Approach for Fraud Detection	Uses graph embeddings to model relationships between users and transactions in a bipartite graph.	Captures complex non-linear relationships	High computational cost

Gaps Identified in Existing Research

The review highlights a persistent, critical gap that the Account GANomaly aims to bridge: the failure of existing models to adequately address personalized and non-stationary anomaly detection.

1. **Lack of Personalized Normality Modeling (The Central Gap):** No surveyed paper explicitly defines a model that learns and enforces an individual user's

baseline of normality. Certain models define "normal" as the average statistical or temporal pattern across the entire dataset, failing to account for user-specific behavior (e.g., a high-roller's transactions being normal for them but anomalous for the overall population). "Anomaly Detection in Networks with Application to Financial Transaction Networks"[4] excels at identifying structural anomalies like rings and heavy paths in networks, but this framework flags suspicious groups, not an individual transaction relative to its owner's history. The focus is structural and not behavioral.

2. **Inadequate Multi-Scale Feature Fusion for Personalization:** While the necessity for combining temporal features is established, neither provides a dedicated, explicit module to fuse short-term sequential signals and long-term aggregate statistics into a single, comprehensive user profile embedding (u). The proposed Hybrid User Encoder directly addresses this missing module.

3. **Need for Adaptive Preprocessing Integration:** "Comparative Study on the Performance of Categorical Variable Encoders in Classification and Regression Tasks " establishes the critical link between the choice of encoder (e.g., One-Hot) and model type (ATI models like GANomaly). This finding highlights a practical implementation gap which the system design must explicitly integrate and justify the chosen categorical encoding technique to ensure optimal performance, a step often left ambiguous in other complex model architectures.

4. **Integration of Structural and Behavioral Learning:** The model architecture needs to find a way to incorporate insights from network-based detection by leveraging structural/relational features within the user context, enhancing the traditional time-series and distributional checks.

2.4 Summary of Chapter

This chapter focuses on anomaly detection, fraud detection, and machine learning techniques for identifying abnormal patterns in data. The GANomaly paper proposes a semi-supervised anomaly detection model using a GAN with an encoder–decoder–encoder architecture to learn normal data patterns and detect anomalies through reconstruction

errors. The ALGAN paper introduces a GAN combined with an Adjusted-LSTM with attention mechanisms to detect anomalies in time-series data by capturing long-term temporal dependencies. The GAN-based Fraud Detection paper applies Generative Adversarial Networks to learn normal transaction behavior and generate synthetic data to improve fraud detection accuracy in financial systems. The XGBoost paper presents a scalable and efficient gradient boosting algorithm that improves prediction performance through optimized tree learning, regularization, and parallel processing for large datasets. Finally, the FraudNE paper proposes a deep joint network embedding approach that models user–item interactions as a bipartite graph and uses autoencoders to learn latent representations, enabling the detection of fraudulent users and suspicious item groups through clustering without needing predefined fraud patterns.

Chapter 3

System Design

The core of this project is the Account GANomaly system, a novel, user-conditioned Generative Adversarial Network (GANomaly) designed for personalized anomaly scoring. The design is driven by the formal problem of training an unsupervised model that accurately reconstructs transaction features only if they align with the user's learned profile (u), making the reconstruction error a direct measure of personalized anomaly.

3.1 Overall Architecture

The proposed system, Account-GANomaly, is a hybrid framework designed to establish a personalized baseline of "normality" for every individual account.. The architecture integrates a user-conditioned Generative Adversarial Network (GAN) with an XGBoost classifier.. It functions by processing both short-term transaction sequences and long-term behavioral statistics into a robust user embedding, which then conditions the GAN to identify deviations unique to a specific user's profile.

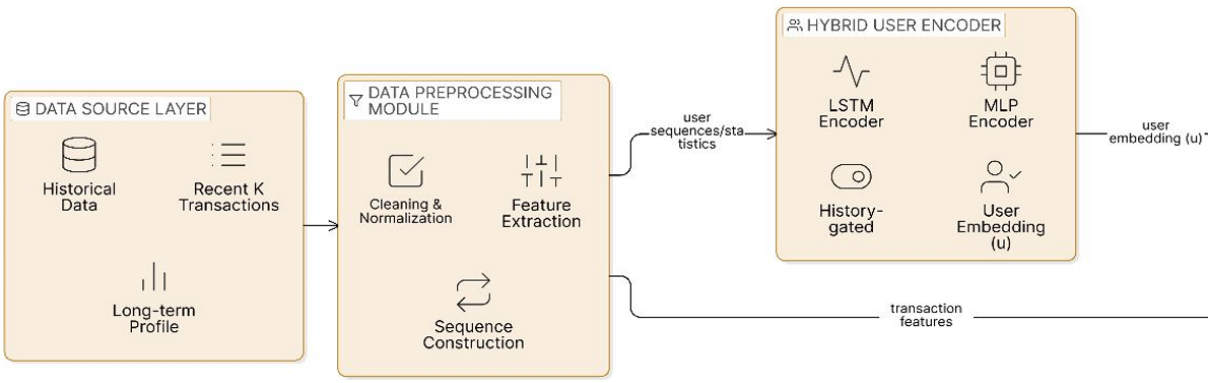

Figure 3.1: Architecture Diagram Part-1

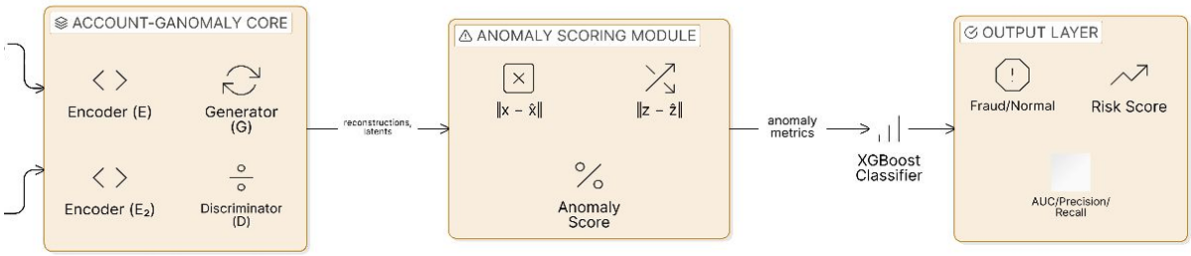

Figure 3.2: Architecture Diagram Part-2

3.2 Component Design

The system architecture of Account-GANomaly is modular, consisting of a hybrid user encoder, a conditioned generative network, and a supervised refinement layer. Each component is designed to work in synergy to provide personalized fraud detection.

3.2.1 Hybrid User Encoder

The Hybrid User Encoder is the personalization engine of the system. It processes raw transaction data to create a compressed 128-dimensional user embedding.

- **Short-term Encoder (LSTM):** This component processes a sequence of the last N transactions (e.g., amount, time-delta, merchant category). By using Long Short-Term Memory (LSTM) units, it captures the temporal dependencies and evolving patterns in a user's recent spending habits.
- **Long-term Encoder (MLP):** This component takes aggregated profile statistics, such as average monthly spend, transaction frequency, and historical variance. A Multi-Layer Perceptron (MLP) compresses these static features into a behavioral profile.

3.2.2 History-Based Gating Mechanism

To solve the "cold-start" problem (where new users have no history), a gating mechanism is implemented.

$$E_{final} = \sigma(W_g \cdot count + b_g) \odot E_{user} \tag{3.1}$$

Where:

- E_{user} is the combined output of the hybrid encoder.
- *count* is the number of historical transactions for that account.
- σ is a sigmoid function that scales the influence of the user embedding based on data reliability.

3.2.3 User-Conditioned GANomaly

The core detection engine follows the GANomaly framework but is uniquely conditioned on the user embedding.

- **Generator (G):** An encoder-decoder network (G_{enc} and G_{dec}) that attempts to reconstruct the input transaction. It is conditioned by concatenating the user embedding to the latent representation.
- **Discriminator (D):** Evaluates the authenticity of the reconstructed transaction.
- **Evaluation Encoder (E):** A secondary encoder that maps the reconstructed transaction back to latent space. The distance between the original latent vector (z) and the reconstructed latent vector ($\hat{z}$) forms the basis of the anomaly score.

3.2.4 Anomaly Scoring and Decision Module

This component calculates the final fraud probability.

1. **Reconstruction Score:** Measures the pixel-wise or feature-wise difference between input x and output $\hat{x}$.
2. **Latent Score:** Measures the distance between z and $\hat{z}$ in the latent space.
3. **XGBoost Classifier:** The scores, along with the user embeddings, are fed into an XGBoost model. This supervised layer learns the final decision boundary to distinguish between high-variance legitimate behavior and actual fraudulent anomalies.

3.3 Module Division

The Account-GANomaly system is divided into five primary modules, each handling a specific stage of the pipeline from raw data ingestion to final fraud classification.

3.3.1 Module 1: Data Preprocessing and Feature Engineering

This module is responsible for preparing the raw financial data for the neural network.

- **Sequence Construction:** Groups the most recent N transactions for each user into temporal sequences.
- **Profile Aggregation:** Calculates long-term statistics such as mean transaction value and frequency per account.
- **Normalization:** Scales numerical features (e.g., amount) and encodes categorical variables (e.g., merchant ID) to ensure uniform input distribution.

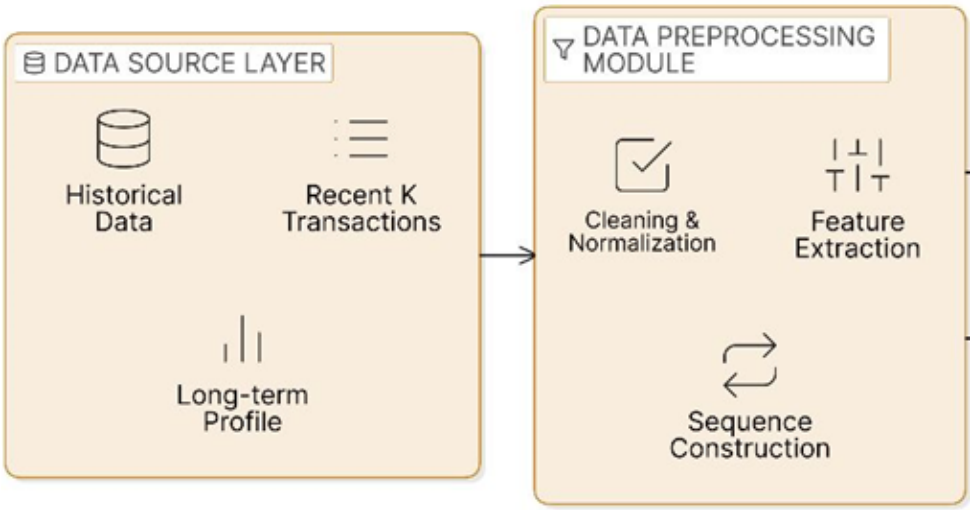

Figure 3.3: Input Layer and Data Preprocessing

3.3.2 Module 2: Gated Hybrid User Encoder

This module generates the personalized context for the detection engine.

- **LSTM/MLP Fusion:** Pass short-term sequences through an LSTM and long-term profiles through an MLP.
- **Embedding Generation:** Concatenates the outputs into a 128-dimensional user vector.
- **History Gating:** Applies a sigmoid-based gate to weigh the importance of the user profile based on the account's transaction history count.

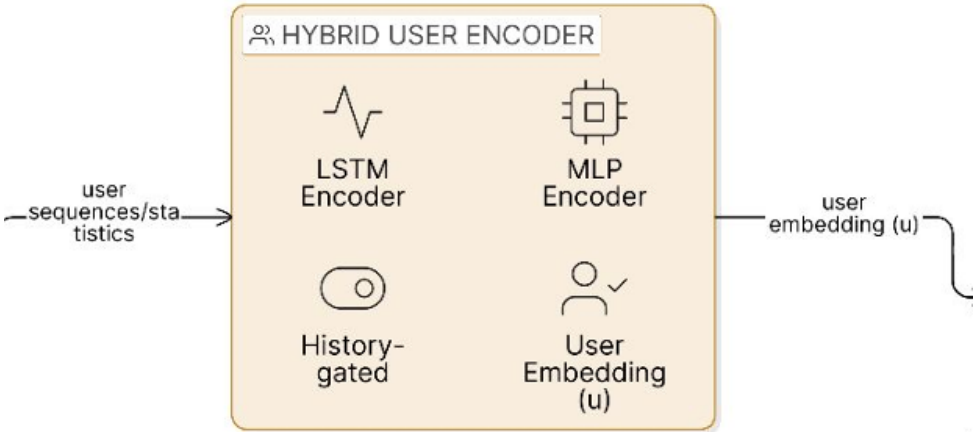

Figure 3.4: Hybrid User Encoder

3.3.3 Module 3: User-Conditioned GANomaly Learner

This is the core unsupervised learning module.

- **Conditional Generation:** The generator (G) takes the current transaction and the user embedding to create a reconstructed version.
- **Adversarial Training:** The discriminator (D) attempts to distinguish between real and reconstructed transactions, forcing the generator to learn user-specific "normality."
- **Latent Mapping:** Maps transactions to a lower-dimensional manifold to capture the underlying behavioral distribution.

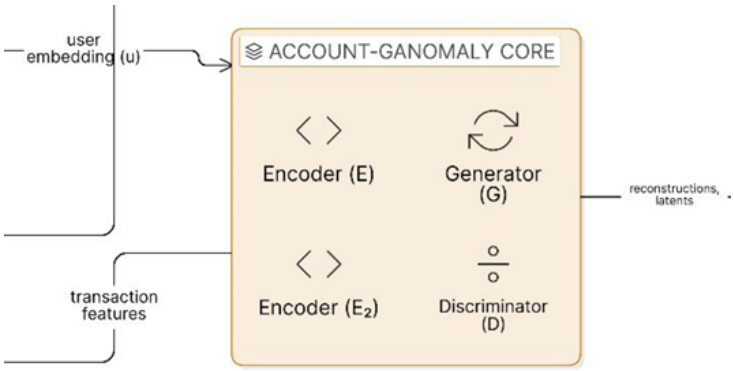

Figure 3.5: User-Conditioned GANomaly

3.3.4 Module 4: Anomaly Scoring and Evaluation

This module quantifies how much a transaction deviates from the user's learned baseline.

- **Error Calculation:** Computes the L_2 distance for reconstruction error and latent space inconsistency.
- **Feature Attribution:** Identifies which specific features (e.g., an unusual amount or location) contributed most to the anomaly score.

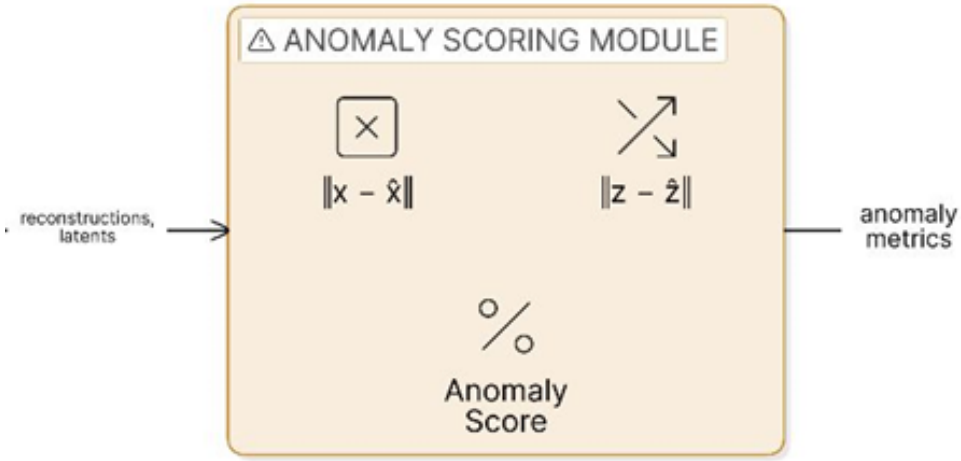

Figure 3.6: Anomaly Scoring and Loss Module

3.3.5 Module 5: XGBoost Supervised Decision Layer

The final module provides the classification outcome.

- **Classifier Integration:** Takes the unsupervised anomaly scores and user embeddings as inputs.
- **Decision Boundary:** Uses a Gradient Boosting Decision Tree (GBDT) framework to output a final fraud probability, significantly reducing false positives by learning from historical labels.

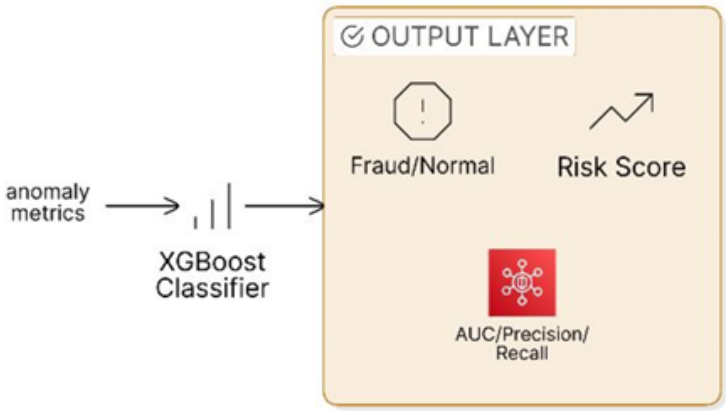

Figure 3.7: XGBoost Supervised Decision Layer

3.4 Tools and Technologies

The development of the Account-GANomaly system utilizes a suite of modern machine learning libraries and hardware acceleration to handle the complex computations required by the GAN architecture and the large-scale financial datasets.

3.4.1 Datasets Used

To evaluate the effectiveness of the proposed *Account-GANomaly* framework, two datasets were used in this project: the IEEE-CIS Fraud Detection dataset and a synthetic transaction dataset.

- **IEEE-CIS Fraud Detection Dataset:** The IEEE-CIS Fraud Detection dataset is a widely used benchmark dataset for financial fraud detection research. It contains approximately 590,000 transaction records with more than 430 anonymized features describing transaction details, user information, and device characteristics.The dataset is highly imbalanced, with a fraud rate of approximately 3.5%.

- **Synthetic Transaction Dataset:** In addition to the real-world dataset, a synthetic transaction dataset was generated to model realistic behavioral patterns of users over time. This dataset contains approximately 2.5 million transactions generated from around 20,000 users.

3.4.2 Hardware Requirements

- **Processor:** High-performance multi-core CPU (Intel Core i7 or equivalent) for data preprocessing and XGBoost training.
- **Memory:** Minimum 16 GB RAM to handle large-scale transaction datasets in-memory.
- **GPU/TPU:** NVIDIA GPU with CUDA support or Google TPU (Tensor Processing Unit) for accelerating the training of the LSTM and GANomaly neural networks.

3.4.3 Software and Frameworks

- **Operating System:** Linux-based environment (Ubuntu) or Windows with WSL2.
- **Development Environment:** Google Colab and Jupyter Notebooks for iterative development and experimentation.
- **Programming Language:** Python 3.x, selected for its extensive ecosystem of machine learning and data science libraries.

3.4.4 Libraries and Dependencies

- **PyTorch:** The primary deep learning framework used to implement the custom Generator, Discriminator, and Hybrid User Encoder architectures.
- **XGBoost:** A scalable gradient boosting library used for the supervised decision layer to provide high-performance classification.
- **Pandas & NumPy:** Essential for data manipulation, cleaning, and numerical computations during the preprocessing of the IEEE-CIS and synthetic datasets.
- **Scikit-Learn:** Used for evaluating model performance metrics (AUC-ROC, Precision, Recall) and for standard data splitting (Train/Test/Validation).
- **Matplotlib & Seaborn:** Utilized for generating visualizations, including the anomaly score distribution and model loss curves.

3.5 Work breakdown and Responsibilities

The project was executed by a team of four members, with tasks divided according to the core functional areas of the system: data engineering, neural network development, classifier optimization, and system integration.

Aaron James [U2203003] : Handled comprehensive data preprocessing and feature engineering for the IEEE-CIS and Synthetic datasets. Developed the training pipeline and implemented the XGBoost supervised layer to establish the final fraud classification decision boundary.

Abhishek S [U2203011] : Managed the critical integration of unsupervised anomaly scores and latent representations into the hybrid framework. Ensured seamless data flow between the GANomaly reconstruction module and the secondary classifier for improved system stability.

Augustin Vincent [U2203061] : Designed the core GANomaly architecture, including the user-conditioned sub-networks and the hybrid LSTM-MLP encoder. Optimized the adversarial training process and performed extensive hyperparameter tuning to stabilize the model.

Bosco Reji Jose [U2203069] : Led the evaluation phase by performing stratified analysis using Precision, Recall, F1-score, and AUC. Implemented the history-gated evaluation and conducted rigorous threshold tuning to balance detection sensitivity and false positive rates.

3.6 Key Deliverables

The following items represent the core outcomes and technical artifacts delivered at the conclusion of Phase 2 for the Account-GANomaly project:

- **Personalized User Embedding Module:** A trained dual-stream encoder (LSTM + MLP) capable of generating 128-dimensional behavioral vectors for individual bank accounts.

- **User-Conditioned GANomaly Model:** A functional PyTorch implementation of the generative adversarial network, optimized to reconstruct transactions based on user-specific latent context.

- **Integrated Hybrid Pipeline:** A complete end-to-end detection pipeline that seamlessly links the unsupervised GANomaly output with a supervised XGBoost classifier.
- **Synthetic Transaction Generator:** A robust data simulation tool, modeled after the IEEE-CIS dataset, used for stress-testing the system's ability to handle extreme data sparsity.
- **Performance Evaluation Suite:** A comprehensive set of evaluation results, including AUC-ROC curves, precision-recall metrics, and feature attribution reports demonstrating a significant reduction in false positives.
- **Project Documentation:** A detailed technical report and presentation detailing the system architecture, mathematical modeling, and comparative analysis against traditional anomaly detection methods.

3.7 Project Timeline

The timeline was structured to ensure a seamless transition from the theoretical research of Phase 1 to the rigorous algorithmic development and performance validation of Phase 2. All planned milestones have been successfully achieved within the projected deadlines, with the system currently in the final documentation and optimization phase.

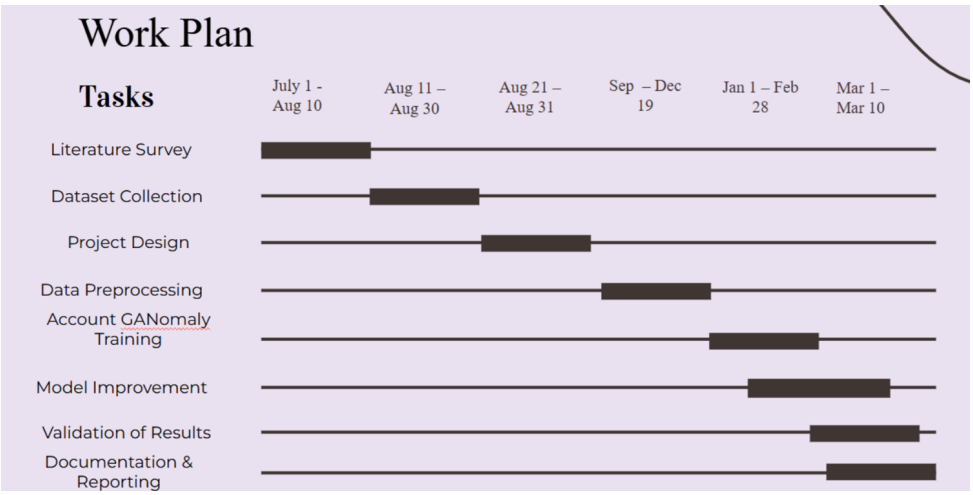

Figure 3.8: Gantt Chart

3.8 Summary of chapter

In this chapter, the design and architectural framework of the Account-GANomaly system were discussed in detail. The chapter outlined the transition from raw financial data through a hybrid encoding process—utilizing both LSTM and MLP architectures—to generate personalized 128-dimensional user embeddings. It further detailed the core detection engine, which leverages a user-conditioned GANomaly model to establish individual behavioral baselines, followed by a supervised XGBoost layer for final decision refinement. Additionally, the chapter defined the modular division of the software, the hardware and software requirements, and the distribution of responsibilities among the team members. By integrating these components, the system establishes a robust pipeline for personalized anomaly detection, the implementation of which will be explored in the subsequent chapters.

Chapter 4

Implementation

This chapter describes the practical implementation of the proposed Account-GANomaly framework for personalized anomaly detection in financial transactions. The goal of the implementation is to develop a hybrid system capable of learning individual user behavior patterns and identifying fraudulent transactions by detecting deviations from these personalized baselines. The system integrates both unsupervised deep learning techniques and supervised machine learning methods to improve fraud detection accuracy while reducing false positives.

4.1 Data Collection and Dataset Preparation

The implementation begins with collecting and preparing the datasets required for training and evaluating the fraud detection model. The project uses two primary datasets: the IEEE-CIS Fraud Detection dataset and a synthetic transaction dataset generated for behavioral modeling. The IEEE dataset contains approximately 590,000 financial transactions with more than 430 anonymized features, while the synthetic dataset includes 2.5 million transactions from around 20,000 users. These datasets provide information such as transaction amount, time intervals, merchant categories, and behavioral patterns, which are essential for learning user-specific transaction profiles.

4.2 Data Preprocessing and Feature Engineering

Before training the models, the transaction data is cleaned and transformed into a suitable format. Missing values and inconsistencies in the dataset are handled using normalization and feature scaling techniques. Feature engineering is performed to generate meaningful attributes such as transaction frequency, time gaps between transactions, account balances, and merchant categories. Additionally, short-term transaction sequences are con-

structed for each user to capture temporal behavior patterns. These processed features are then normalized to ensure stable training of neural network models.

4.3 Hybrid User Encoder for Personalized Embeddings

To capture personalized transaction behavior, a hybrid user encoder is implemented. This module combines two neural network architectures:

- **LSTM (Long Short-Term Memory):** Used for learning short-term sequential transaction patterns.
- **MLP (Multilayer Perceptron):** Used for learning long-term aggregated statistical features of user behavior.

The outputs of these two networks are combined to generate a user embedding, which represents the behavioral profile of an individual account. A history-based gating mechanism is also introduced to adjust the influence of the embedding based on the number of available transactions, thereby addressing the cold-start problem for new users.

4.4 User-Conditioned GANomaly Model

The core anomaly detection model is implemented using a GANomaly architecture, which consists of three main components:

- **Encoder (E):** Converts the transaction features into a latent representation.
- **Generator (G):** Reconstructs the transaction data based on the latent representation and user embedding.
- **Discriminator (D):** Differentiates between real and reconstructed samples.

The model is conditioned on the user embedding, ensuring that the system learns a personalized baseline of normal behavior. During training, the GAN learns to reconstruct legitimate transactions accurately. When a fraudulent or abnormal transaction occurs, the reconstruction error becomes significantly higher, indicating an anomaly.

4.5 Anomaly Scoring Mechanism

After the reconstruction process, an anomaly score is calculated to quantify how much a transaction deviates from the learned normal behavior. The anomaly score is computed using a combination of two metrics:

- **Reconstruction Error:** The difference between the original transaction and the reconstructed transaction.
- **Latent Representation Difference:** The difference between the encoded latent vectors before and after reconstruction.

The final anomaly score is calculated as a weighted combination of these two values. Transactions with higher anomaly scores are considered suspicious and are passed to the next stage for classification.

4.6 XGBoost Supervised Classification Layer

To improve detection accuracy, the anomaly scores and learned feature representations are fed into a supervised XGBoost classifier. XGBoost is an efficient gradient boosting algorithm that can capture complex non-linear relationships in the data. It learns from labeled transaction data to distinguish between fraudulent and legitimate transactions. This hybrid approach combines the strengths of unsupervised anomaly detection with supervised classification, significantly improving fraud detection performance.

4.7 Model Training and Evaluation

The model is trained using Python with PyTorch for the deep learning components and XGBoost for classification. During training, the GANomaly network learns normal transaction patterns while the XGBoost classifier learns decision boundaries for fraud detection. The system is evaluated using performance metrics such as:

- Accuracy
- Precision
- Recall

- F1-score
- Area Under the ROC Curve (AUC)

Experimental results show that the hybrid model achieves strong performance, with the XGBoost decision layer reaching an AUC score of approximately 0.81, demonstrating its effectiveness in identifying fraudulent transactions.

4.8 Summary of Chapter

The implementation chapter describes the practical development of the proposed *Account-GANomaly* framework for personalized fraud detection in financial transactions. The system begins with collecting and preparing datasets, including the IEEE-CIS Fraud Detection dataset and a synthetic transaction dataset, which provide detailed transaction attributes and behavioral features for model training. The data is then preprocessed through cleaning, normalization, and feature engineering to generate meaningful attributes such as transaction frequency, time intervals, and behavioral patterns. A hybrid user encoder is implemented to generate personalized user embeddings by combining short-term sequential behavior using Long Short-Term Memory (LSTM) networks and long-term statistical features using Multilayer Perceptron (MLP) networks.

These personalized embeddings are integrated into a user-conditioned GANomaly model that learns the normal transaction behavior for each individual user and detects anomalies based on reconstruction error and latent representation differences. The generated anomaly scores and feature representations are then provided to an XGBoost classifier, which performs supervised classification to distinguish between fraudulent and legitimate transactions. The system is implemented using Python with PyTorch for deep learning components and XGBoost for classification. Performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). The experimental results demonstrate that the hybrid approach effectively captures complex behavioral patterns and significantly improves fraud detection accuracy while reducing false positive rates.

Chapter 5

Results and Discussions

This section presents the experimental results of the proposed *Account-GANomaly* framework and analyzes its performance in detecting fraudulent financial transactions. The model was evaluated using two datasets: the IEEE-CIS Fraud Detection dataset and a synthetic transaction dataset designed to capture behavioral transaction patterns. Performance was measured using standard classification metrics such as Accuracy, Precision, Recall, F1-score, and Area Under the ROC Curve (AUC). These metrics provide a comprehensive evaluation of the model's ability to detect fraudulent transactions while minimizing false positives.

5.1 Results on IEEE-CIS Dataset

The IEEE-CIS dataset contains highly imbalanced transaction data, where the fraud rate is approximately 3.5%. The results demonstrate that the proposed hybrid model effectively learns user-specific behavioral patterns and improves fraud detection accuracy. The GANomaly component generates anomaly-aware representations, which are then classified by the XGBoost decision layer. The final XGBoost model achieved an AUC score of approximately 0.81 on the validation dataset, indicating strong performance in distinguishing between fraudulent and legitimate transactions.

The results show that incorporating personalized user embeddings significantly improves fraud detection performance. Compared with global models that treat all users equally, the personalized approach reduces false positives by adapting to individual transaction patterns.

Results - IEEE Dataset

Regime	gan_val_auc	xgb_holdout_auc	xgb_val_auc	xgb_auc_Q1_lowest_history	xgb_auc_Q2	xgb_auc_Q3	xgb_auc_Q4_highest_history
Full Model	0.6200	0.7907	0.8102	0.7987	0.8081	0.7657	0.8122
Implicit Personalization	0.6616	0.8036	0.7866	0.7701	0.7721	0.7774	0.7878
Fully Global	N/A	0.8047	0.7837	0.7701	0.7721	0.7774	0.7878

Figure 5.1: Performance results of the proposed model on the IEEE-CIS Fraud Detection dataset.

5.2 Results on Synthetic Dataset

The synthetic dataset was designed to simulate real-world behavioral transaction patterns across approximately 20,000 users and 2.5 million transactions. This dataset allows the evaluation of the model's ability to detect burst fraud patterns and abnormal behavioral sequences.

Results - Synthetic Dataset

Regime	gan_val_auc	xgb_holdout_auc	xgb_val_auc	xgb_auc_Q1_lowest_history	xgb_auc_Q2	xgb_auc_Q3	xgb_auc_Q4_highest_history
Full Model	0.6780	0.8037	0.8123	0.8420	0.8301	0.7967	0.7977
Implicit Personalization	0.5971	0.7247	0.7251	0.7393	0.7194	0.7214	0.7247
Fully Global	N/A	0.7295	0.7282	0.7363	0.7232	0.7283	0.7281

Figure 5.2: Performance results of the proposed model on the synthetic transaction dataset.

Experimental results indicate that the full personalized model achieved an AUC score

of approximately 0.81, outperforming both the implicit personalization model and the fully global baseline. The results also show improved detection performance across different user history groups, particularly for users with limited transaction history.

5.3 Discussions

The experimental analysis highlights the importance of incorporating user-specific behavioral context into fraud detection systems. Traditional anomaly detection methods rely on global thresholds that may incorrectly flag legitimate but unusual transactions as fraudulent. In contrast, the proposed system evaluates whether a transaction is suspicious relative to the behavior of a specific user profile.

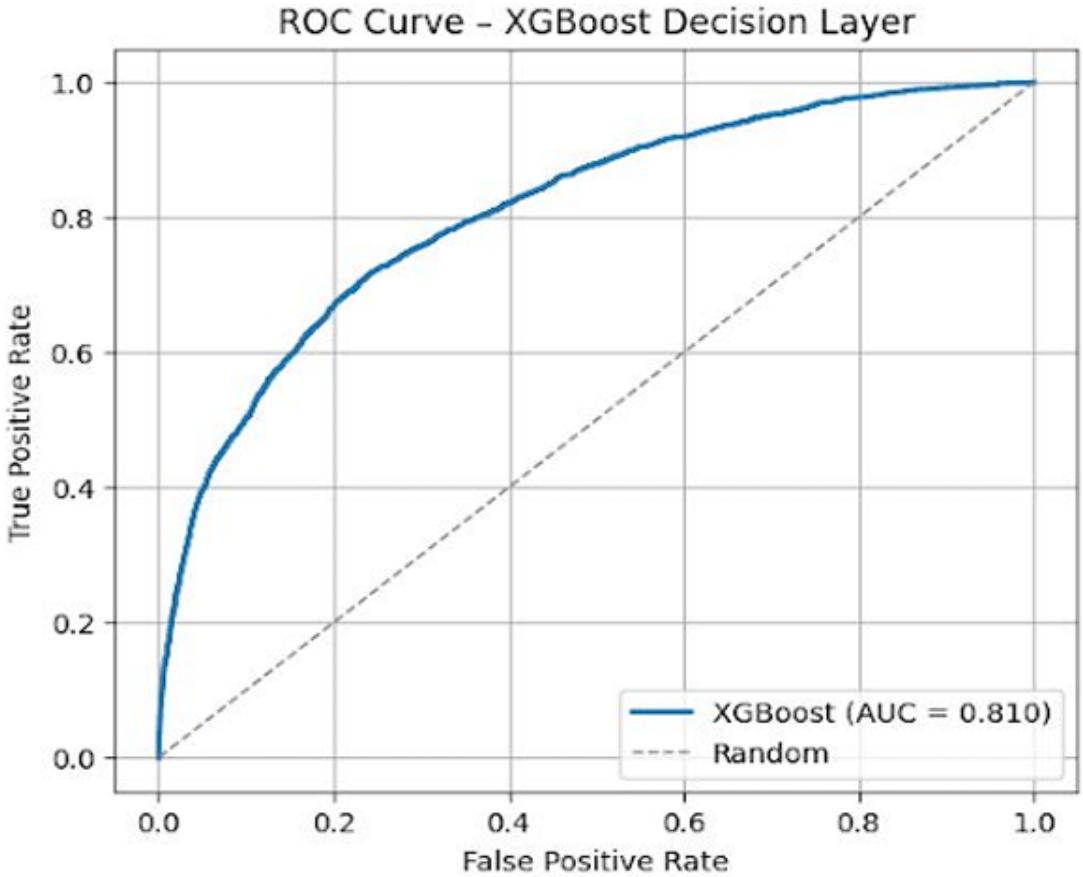

Figure 5.3: Comparison of fraud detection performance across different model configurations.

The results demonstrate three key insights:

- Personalized embeddings improve anomaly detection by aligning the model with

individual behavioral patterns.

- The hybrid architecture combining GANomaly and XGBoost provides stronger performance than using unsupervised methods alone.
- The history-gated encoder effectively mitigates the cold-start problem for users with limited transaction history.

Overall, the proposed system successfully improves fraud detection performance while maintaining stability in highly imbalanced financial datasets.

5.4 Summary of Chapter

This chapter presented the experimental evaluation of the proposed *Account-GANomaly* framework for personalized fraud detection in financial transactions. The system was tested on both the IEEE-CIS Fraud Detection dataset and a synthetic transaction dataset to evaluate its effectiveness in identifying fraudulent behavior under different transaction patterns. The results demonstrated that incorporating personalized user embeddings significantly improves anomaly detection by capturing individual behavioral patterns rather than relying on global transaction characteristics.

The hybrid architecture combining the GANomaly model for unsupervised anomaly representation learning and the XGBoost classifier for supervised decision making achieved strong detection performance, with the XGBoost layer reaching an AUC score of approximately 0.81. The analysis also showed that the proposed history-gated encoder helps mitigate the cold-start problem for users with limited transaction history. Overall, the experimental results confirm that the proposed approach enhances fraud detection accuracy, reduces false positives, and provides a robust framework for detecting suspicious financial transactions in modern digital systems.

Chapter 6

Conclusions & Future Scope

6.1 Conclusion

This project presents a personalized anomaly detection framework for financial transactions using a hybrid deep learning approach. The proposed Account-GANomaly architecture combines a user-conditioned Generative Adversarial Network with an XGBoost classifier to detect fraudulent activities based on individual behavioral patterns rather than global transaction rules. By integrating both short-term transaction sequences through LSTM and long-term behavioral statistics through MLP, the system generates robust user embeddings that define a personalized baseline of normal activity. This approach enables the model to identify deviations that are specific to each user profile, thereby improving detection accuracy while reducing false positives that commonly occur in traditional fraud detection systems.

Experimental evaluation on both the IEEE-CIS dataset and synthetic transaction data demonstrates that the hybrid model effectively captures complex behavioral patterns and adapts to varying transaction histories. The combination of unsupervised anomaly representation learning and supervised classification enhances the system's ability to detect suspicious financial behavior while maintaining stability in highly imbalanced datasets. Overall, the project highlights the effectiveness of personalized anomaly detection techniques in strengthening security for modern financial systems and demonstrates how advanced machine learning models can improve fraud detection in large-scale digital economies.

6.2 Future Scope

In the future, the system can be extended to support real-time fraud detection in live banking environments by integrating streaming transaction processing frameworks. The model can also be enhanced by incorporating additional behavioral signals such as location

data, device fingerprints, and network activity to improve contextual understanding of user behavior. Another possible extension is the use of transformer-based architectures for sequential modeling, which may capture longer behavioral dependencies more effectively than LSTM networks. Additionally, adaptive learning mechanisms can be implemented to address concept drift, allowing the system to continuously update user behavior profiles as financial patterns evolve. Finally, integrating the model with blockchain-based verification or secure banking infrastructures could further enhance the reliability and security of financial transaction monitoring systems.

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Appendix A: Presentation

Personalized Anomaly Detection in Financial Transactions

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- Project Objective
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Problem Definition

In today's rapidly evolving digital economy, fraudulent financial transactions have become increasingly sophisticated and pose a significant threat to global economic stability.

Traditional anomaly detection systems often fail to keep pace because the definition of a normal transaction depends entirely on individual user behavior.

This creates a critical gap where standard models cannot effectively distinguish between legitimate high-value activity and actual fraud due to extreme data sparsity and the lack of personalized baselines.

Project Objective

The project intends to develop a user-conditioned Generative Adversarial Network coupled with an XGBoost classifier that learns to establish a personalized baseline of normality for every individual account.

By integrating short-term transaction sequences and long-term aggregated statistics into a robust user embedding the system can accurately identify deviations that are unique to a specific profile.

This approach aims to maximize detection accuracy and reduce false positives by ensuring the model only reconstructs features that align with the learned behavior of a particular user.

Purpose & Need

The purpose of this research is to develop **Account-GANomaly**, a hybrid framework that integrates user-conditioned Generative Adversarial Networks with XGBoost. By learning a personalized "reconstruction baseline" for every account, the project intends to:

- **Personalize Detection:** Create robust user embeddings that combine short-term behavior and long-term statistics.
- **Enhance Accuracy:** Utilize a supervised XGBoost layer to refine the unsupervised GANomaly scores and capture complex non-linear fraud patterns.
- **Reduce False Alarms:** Ensure that legitimate but rare transactions are not flagged as fraud if they align with the learned user profile.

Literature Survey - Paper I

Utilizing GANs for Fraud Detection: Model Training with Synthetic Transaction Data

Methodology - The paper proposes an **AI fraud prevention system** that utilizes Generative Adversarial Networks (GANs) to enhance transaction security, specifically focusing on verifying that users are real humans rather than automated bots.

Advantages -

- Unsupervised Learning
- High Sensitivity to Complex Patterns
- Improved Training Sets
- Enhanced Security

Disadvantages -

- High Data Requirement
- Hyperparameter Sensitivity

Literature Survey - Paper II

GANomaly: Semi-Supervised Anomaly Detection via Adversarial Training

Methodology - The paper introduces a novel deep learning architecture for anomaly detection that utilizes a conditional Generative Adversarial Network (cGAN) to learn the distribution of "normal" data.

Advantages -

- High Performance on Imbalanced Data
- Robustness to Noise
- State-of-the-Art Accuracy

Disadvantages -

- Computational Complexity
- Training Bottlenecks
- Sensitivity to Normal Variation

Literature Survey - Paper III

XGBoost: A Scalable Tree Boosting System

Methodology - XGBoost involves minimizing a regularized objective function that balances a differentiable loss function with a penalty for model complexity to prevent overfitting. To achieve extreme scalability, the system implements a sparsity-aware algorithm for handling missing data, a weighted quantile sketch for approximate split finding and a cache-aware block structure for parallelized tree construction.

Advantages -

- Extreme Scalability
- Sparsity Handling
- Resource Efficiency

Disadvantages -

- Hyperparameter Complexity
- Diminishing Returns on Ensembles

Literature Survey - Paper IV

ALGAN:Time Series Anomaly Detection with Adjusted- LSTM GAN

Methodology - ALGAN involves training a Generative Adversarial Network (GAN) using a novel **Adjusted-LSTM** architecture to learn the distribution of normal time series sequences in an unsupervised manner. Anomalies are then identified by mapping real data to a latent space and calculating an **anomaly score** based on the reconstruction loss between the original sequence and its generated counterpart

Advantages -

- Superior Benchmarking
- Unsupervised Efficiency
- Temporal Sensitivity

Disadvantages -

- Sequential Bottlenecks
- Memory Intensity

Literature Survey - Paper V

FraudNE: Joint Embedding Approach for Fraud Detection

Methodology - The paper proposes **FraudNE**, a joint deep structure embedding approach designed to detect fraudsters in bipartite graphs (users and items) by capturing complex relationships.

Advantages -

- Improved Detection Accuracy
- Robustness to Sparsity
- Captures High Non-linearity

Disadvantages -

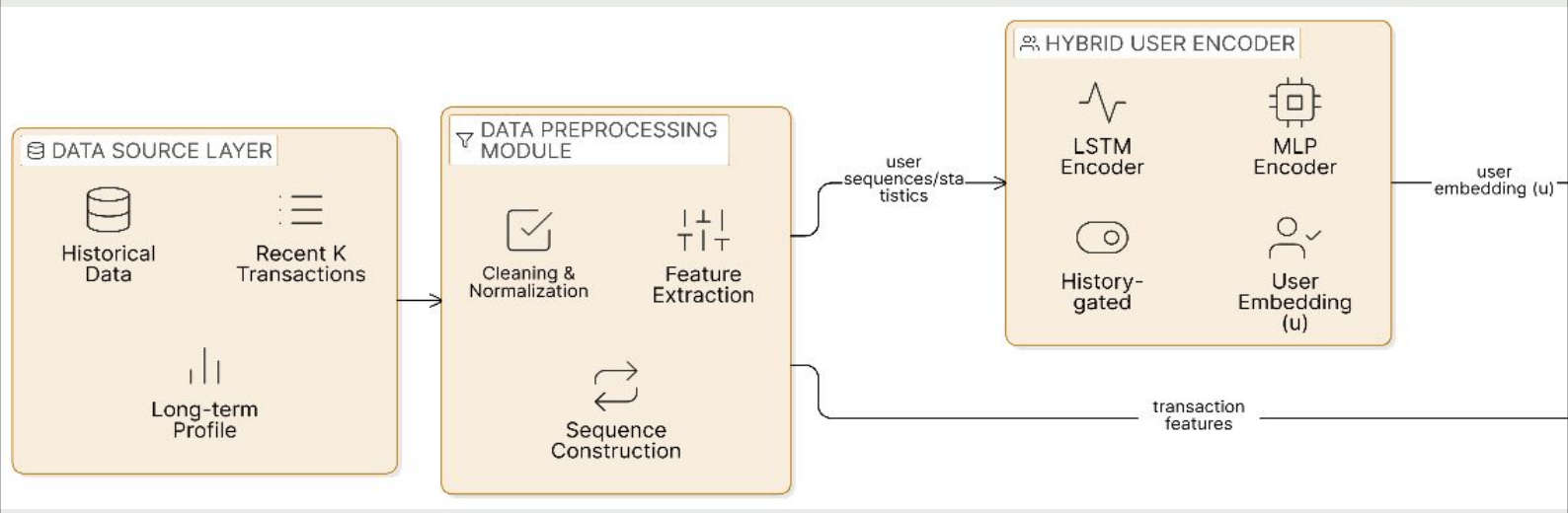
- High Computational Cost
- Sensitivity to Graph Noise
- Cold-Start Limitations

Proposed Method

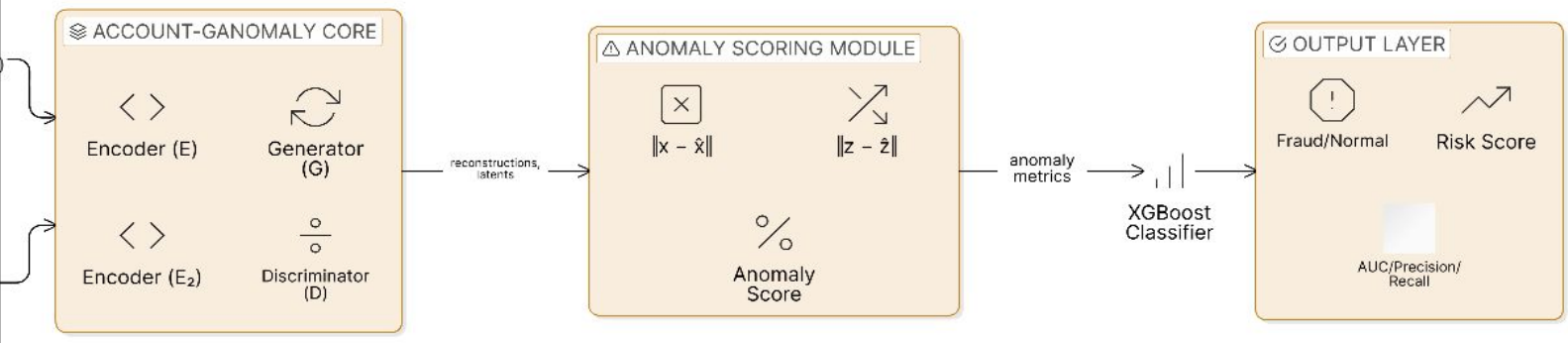
Methodology: Account-GANomaly Architecture

- **Hybrid User Encoder:** The system employs a dual-stream encoder that combines **Long Short-Term Memory (LSTM)** networks for short-term transaction sequences and **Multilayer Perceptrons (MLP)** for long-term aggregated profile statistics.
- **History-Based Gating:** To solve the cold-start problem the model utilizes a gating function that adjusts the weight of the user embedding based on the number of available transactions.
- **User-Conditioned GANomaly:** The learned user representation is integrated directly into the Generator and Discriminator to ensure that reconstruction error measures deviations from a personalized baseline of normality.
- **Supervised Hybrid Pipeline:** An unsupervised representation learner captures complex features and then passes them to an **XGBoost classifier** to establish a refined supervised decision boundary for final fraud detection.

Architecture Diagram



Architecture Diagram



Module Description

Module 1: Input Layer and Data Preprocessing

- This module manages the initial feature engineering by preparing the transaction features and building the short-term sequence records for every user. It computes long-term profile statistics to provide a comprehensive behavioral overview and ensures all inputs are normalized for the neural network layers.

Module 2: Hybrid User Encoder (Gated Personalization)

- The personalization module generates a robust user embedding by fusing a short-term LSTM with a long-term MLP. It includes a history-based gating mechanism that controls the influence of the embedding based on the available transaction history to mitigate the cold-start problem.

Module 3: User-Conditioned GANomaly

- This core unsupervised learner utilizes an Encoder-Generator-Encoder architecture to model normal transaction behavior conditioned on the user context. It learns to reconstruct transaction features such that the reconstruction error and latent inconsistency serve as direct measures of personalized anomaly.

Module Description

Module 4: Anomaly Scoring and Loss Module

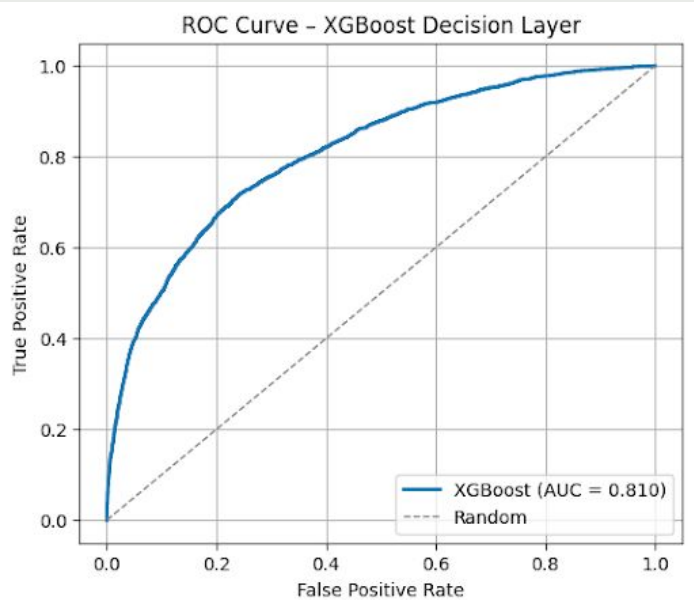
- The scoring module quantifies the deviation from learned normality by balancing the reconstruction error and latent difference between the input and generated outputs. It produces anomaly-aware representations and per-feature error metrics that highlight exactly which transaction attributes are suspicious.

Module 5: XGBoost Supervised Decision Layer

- The final module acts as a supervised classifier that receives the learned representations and anomaly scores from the GANomaly model. It learns non-linear and label-driven fraud patterns to establish a precise decision boundary which significantly improves the detection accuracy over unsupervised methods alone.

Results & Outputs

```
=== XGBoost stratified AUCs ===
0 (cold-start) | n=   417 | XGB AUC=0.787
1–2            | n=   760 | XGB AUC=0.618
3–5            | n=   962 | XGB AUC=0.858
6–10           | n=  1490 | XGB AUC=0.790
10+            | n= 55425 | XGB AUC=0.813
```



Results - IEEE Dataset

Regime	gan_val_auc	xgb_holdout_auc	xgb_val_auc	xgb_auc_Q1_lowest_history	xgb_auc_Q2	xgb_auc_Q3	xgb_auc_Q4_highest_history
Full Model	0.6200	0.7907	0.8102	0.7987	0.8081	0.7657	0.8122
Implicit Personalization	0.6616	0.8036	0.7866	0.7701	0.7721	0.7774	0.7878
Fully Global		0.8047	0.7837	0.7701	0.7721	0.7774	0.7878

Results - Synthetic Dataset

Regime	gan_val_auc	xgb_holdout_auc	xgb_val_auc	xgb_auc_Q1_lowest_history	xgb_auc_Q2	xgb_auc_Q3	xgb_auc_Q4_highest_history
Full Model	0.6780	0.8037	0.8123	0.8420	0.8301	0.7967	0.7977
Implicit Personalization	0.5971	0.7247	0.7251	0.7393	0.7194	0.7214	0.7247
Fully Global		0.7295	0.7282	0.7363	0.7232	0.7283	0.7281

Results Takeaways

1. Core Research Insights

- Context over Performance: Personalization is a context amplifier. It succeeds where behavioral structure is present.
- The "Topology" Effect :
  - Synthetic (Behavioral): High impact on cold-starts and detecting short-term burst fraud.
  - IEEE-CIS (Identity): Benefits from long-term stability; performance scales as user history matures.

2. Industrial Value Proposition

- A Shift in Logic: Moves the question from "Is this transaction suspicious?" to "Is this transaction suspicious for this specific user?"
  - Reduced False Positives: Highly stable users aren't flagged for "global" anomalies that are "local" norms.
  - Adaptive Thresholds: Enables dynamic risk scoring tailored to individual spending patterns.

User embeddings provide a strategic, context-aware layer to anomaly detection. Their value lies in aligning the model architecture with the underlying behavioral signature of the user.

Project Assumptions

- **Consistency of User Profiles:** It is assumed that while financial behavior is non-stationary it still contains underlying latent patterns that can be captured as a stable user embedding.
- **Data Integrity and Availability:** The model assumes that the provided transaction histories are sufficiently accurate to establish a baseline of normality and that enough features exist to distinguish between different user behaviors.
- **Fraud as a Deviation:** A fundamental assumption is that fraudulent transactions will significantly deviate from a user's learned reconstruction profile and thus result in a higher error score than legitimate activities.
- **Stationarity of Normalcy:** While the model adapts to change it assumes that the transition between different behavioral regimes is gradual enough for the conditional GAN to update its internal representation of normality.

Work Breakdown & Responsibility

Aaron James: Handled comprehensive data preprocessing and feature engineering for the IEEE-CIS and Synthetic datasets. Developed the training pipeline and implemented the **XGBoost supervised layer** to establish the final fraud classification decision boundary.

Abhishek S: Managed the critical integration of unsupervised anomaly scores and latent representations into the hybrid framework. Ensured seamless data flow between the GANomaly reconstruction module and the secondary classifier for improved system stability.

Augustin Vincent: Designed the core **GANomaly architecture**, including the user-conditioned sub-networks and the hybrid LSTM-MLP encoder. Optimized the adversarial training process and performed extensive hyperparameter tuning to stabilize the model.

Bosco Reji Jose: Led the evaluation phase by performing stratified analysis using **Precision, Recall, F1-score, and AUC**. Implemented the history-gated evaluation and conducted rigorous threshold tuning to balance detection sensitivity and false positive rates.

Hardware & Software Requirements

Hardware Requirements

- Laptop/Desktop (8 GB RAM or higher recommended)
- Stable internet connection
- Secure campus network or cloud environment

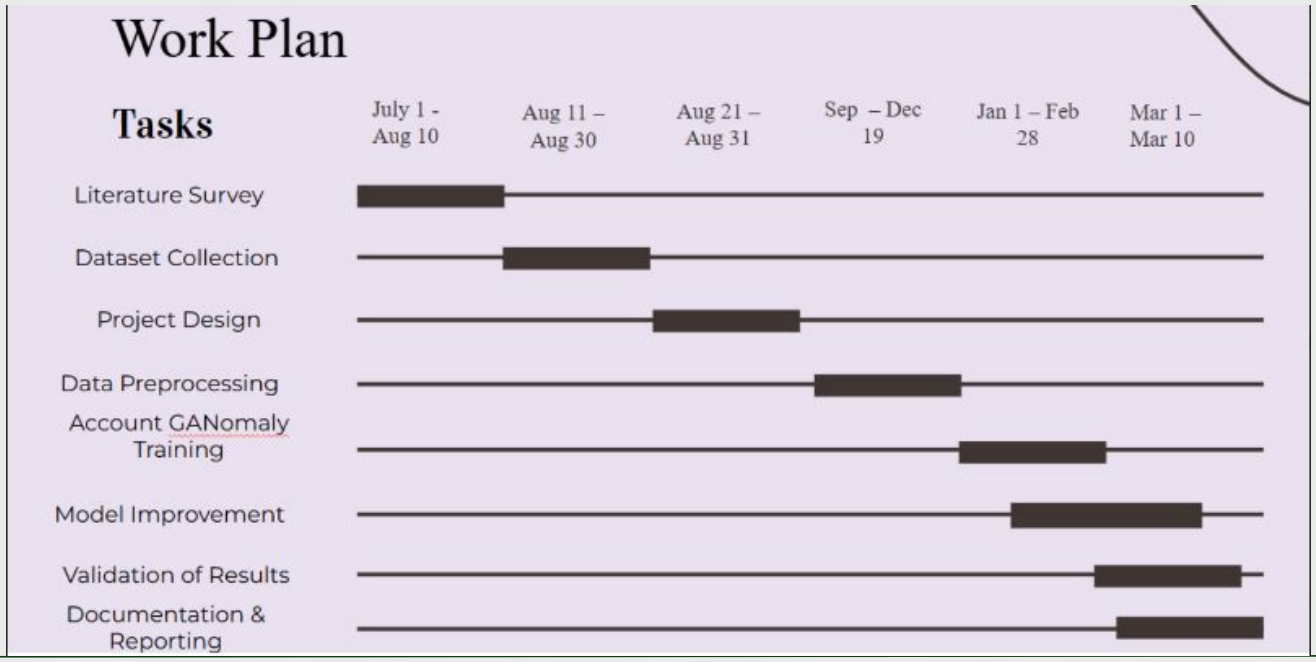
Software Requirements

- Python 3.8
- Datasets Used: IEEE-CIS Fraud Detection dataset and Synthetic transaction data generated for longitudinal profiling
- PyTorch (for GANomaly and LSTM) and XGBoost (for supervised classification)
- VS Code / IDE for development
- GitHub for version control

Datasets Used

Feature	IEEE-CIS Dataset	Synthetic Dataset
Primary Goal	Feature-based fraud detection	Behavioral anomaly & burst detection
Data Scale	~590k Transactions	~2.5M Transactions (20k Users)
User Identity	Proxy-based (Card/ID merge)	Explicit user_id (Clean history)
Fraud Rate	3.5% (Extremely Imbalanced)	1.5% (Controlled Sparsity)
Key Attributes	430+ Anonymized V-features	Time-deltas, Balances, Merchant Cats
Temporal Logic	Limited/Noisy sequences	Structured Bursts (3–7 txns)

Gantt Chart



Risks & Challenges

- **Extreme Data Sparsity:** Many individual accounts have very few transactions which makes it difficult for the LSTM and MLP encoders to generate a truly representative user embedding.
- **Cold-Start Sensitivity:** New users lack the historical data required to establish a personalized baseline and the model may initially produce high false positive rates until the profile matures.
- **Adversarial Training Instability:** Balancing the Generator and Discriminator in a user-conditioned environment is complex and can lead to mode collapse or gradient disappearance during the training phase.
- **Concept Drift:** Financial behaviors change over time due to seasonal trends or life events and the model must be carefully updated to ensure it does not flag legitimate behavioral shifts as fraud.

Conclusion

- **Establishment of Personalized Baselines:** The project successfully demonstrates that conditioning a GANomaly architecture on user-specific embeddings allows for the creation of a highly accurate and personalized baseline of normality.
- **Effective Hybrid Integration:** The combination of unsupervised reconstruction error from the GAN with the supervised precision of XGBoost significantly enhances detection rates while maintaining a low false positive threshold.
- **Resilience to Data Challenges:** Through the implementation of a history-gated encoder the system effectively manages the cold-start problem and addresses the extreme data sparsity inherent in individual financial profiles.
- **Advancement in Fraud Detection:** This research provides a robust framework for real-time anomaly detection that adapts to the non-stationary nature of transaction behavior and ensures superior security in modern digital economies.

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Appendix B: Vision, Mission and Programme Outcomes

Vision, Mission and Programme Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

2. Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for

sustainable development.

3. Design/Development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

4. Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

5. Engineering Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

8. Individual and Collaborative Team work:Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

10. Project management and finance: Apply knowledge and understanding of en-

gineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills
The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills
The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills
The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Appendix C: Course Outcomes

Course Outcomes (CO)

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND PSOs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO–PO MAPPING

Mapping	Level	Justification
101003/CS822U.CO1-PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/CS822U.CO1-PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/CS822U.CO1-PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/CS822U.CO1-PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/CS822U.CO1-PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/CS822U.CO1-PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/CS822U.CO1-PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/CS822U.CO1-PO9	Low	Requires communication of results and findings through reports or presentations.
101003/CS822U.CO1-PO10	Low	Involves basic planning and execution of tasks while implementing solutions.

Mapping	Level	Justification
101003/CS822U.CO1-PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/CS822U.CO1-PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/CS822U.CO1-PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/CS822U.CO1-PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/CS822U.CO2-PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/CS822U.CO2-PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/CS822U.CO2-PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/CS822U.CO2-PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/CS822U.CO2-PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/CS822U.CO2-PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/CS822U.CO2-PO8	Low	Development activities may require collaboration among team members.
101003/CS822U.CO2-PO9	Low	Involves communicating design ideas and development outcomes.
101003/CS822U.CO2-PO10	Low	Requires basic planning and coordination during development tasks.
101003/CS822U.CO2-PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/CS822U.CO2-PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/CS822U.CO2-PSO2	Medium	Uses programming and software development practices to implement solutions.

Mapping	Level	Justification
101003/CS822U.CO3-PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/CS822U.CO3-PO9	Medium	Requires effective communication among team members during collaboration.
101003/CS822U.CO3-PO10	Medium	Team leadership involves project coordination and task management.
101003/CS822U.CO3-PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/CS822U.CO3-PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/CS822U.CO4-PO2	Low	Requires analysis and planning before executing project tasks.
101003/CS822U.CO4-PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/CS822U.CO4-PO5	High	Requires effective use of modern tools and resources during project execution.
101003/CS822U.CO4-PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/CS822U.CO4-PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/CS822U.CO4-PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/CS822U.CO4-PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/CS822U.CO4-PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/CS822U.CO4-PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/CS822U.CO4-PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/CS822U.CO4-PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/CS822U.CO5-PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.

Mapping	Level	Justification
101003/CS822U.CO5-PO2	High	Requires deep problem analysis and identification of research gaps.
101003/CS822U.CO5-PO3	High	Proposes innovative design solutions to address identified gaps.
101003/CS822U.CO5-PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/CS822U.CO5-PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/CS822U.CO5-PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/CS822U.CO5-PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/CS822U.CO5-PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/CS822U.CO6-PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/CS822U.CO6-PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/CS822U.CO6-PO7	Medium	Requires responsible and ethical communication of technical results.
101003/CS822U.CO6-PO8	Medium	Communication supports collaboration within team environments.
101003/CS822U.CO6-PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/CS822U.CO6-PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/CS822U.CO6-PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/CS822U.CO6-PSO3	High	Communication of innovative computer science solutions and research outcomes.